# Package 'asaData'

November 6, 2026

**Type** Package

**Title** Download Datasets from 'Applied Survival Analysis'

**Version** 0.2.1

**Description** An unofficial companion package to the first and second editions of 'Applied Survival Analysis: Regression Modeling of Time to Event Data' by Hosmer and Lemeshow (1999) <ISBN:9780471154105>, and Hosmer, Lemeshow, and May (2008) <doi:10.1002/9780470258019>. Downloads and imports 16 datasets from Wiley companion archives, with verified UCLA alternatives for six datasets and offline imports from local files, archives, or folders. Preserves original numeric coding and dates, checks source files, and documents variables and edition differences. No dataset records or original codebooks are included.

**License** GPL-3

**Encoding** UTF-8

**Language** en-GB

**Depends** R (>= 3.5.0)

**Imports** foreign

**Suggests** survival,
knitr,
rmarkdown

**VignetteBuilder** knitr

**URL** https://github.com/graemeleehickey/asaData

**BugReports** https://github.com/graemeleehickey/asaData/issues

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## Contents


asaData-package . . . . . 2
actg320 . . . . . 3
actg320ncc . . . . . 5
asa_datasets . . . . . 6
bpd . . . . . 6
comprisk . . . . . 8

cvdrisk . . . . . 9
download_asa . . . . . 10
frtcs . . . . . 11
gbcs . . . . . 13
grace1000 . . . . . 14
hmohiv . . . . . 16
read_asa . . . . . 17
read_asa_all . . . . . 18
recur . . . . . 19
uis . . . . . 20
whas . . . . . 21
whas100 . . . . . 23
whas500 . . . . . 24
whasncc_e1 . . . . . 25
whasncc_e2 . . . . . 27

**Index** **29**

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`asaData-package` *Download Datasets from Applied Survival Analysis*

---

## Description

An unofficial companion package to the first and second editions of Applied Survival Analysis: Regression Modeling of Time to Event Data.

## Details

Use `asa_datasets` to list the 16 supported datasets and `download_asa` to download and import one. Each help page identifies the book edition and describes the variables, category codes, units, and endpoints. The shared `recur` and `uis` datasets follow the second-edition definitions. The different WHAS nested case-control samples are available as `whasncc_e1` and `whasncc_e2`.

Variable names are lower case, category codes are numeric, calendar dates are `Date` objects, and missing-value dots are represented by `NA`. Source row order and numeric values are preserved.

## Source files

No dataset records or original codebooks are included in the package. Downloads prefer Wiley, with verified UCLA alternatives for six datasets, and are cached for the current R session by default. Use `read_asa` to import one dataset from a local file, ZIP archive, or folder, or `read_asa_all` to import all available datasets together. Supplying both Wiley archives to `read_asa_all()` returns a named list of all 16 datasets without internet access. Every import checks the source checksum, dimensions, missing-value counts, and variable types.

## Licensing

Package code and original documentation are distributed under the GNU General Public License, version 3 (GPL-3). Downloaded datasets and codebooks retain their existing rights and are subject to the source terms.

## Author(s)

Graeme L. Hickey

## Source

Wiley companion data site.

UCLA companion page: examples and alternative data files.

## See Also

`download_asa`, `read_asa`, `asa_datasets`

`vignette("survival-examples", package = "asaData")` for analyses using the **survival** package.

## Examples

```
asa_datasets()
## Not run:
whas500 <- download_asa("whas500")
head(whas500)

## End(Not run)
```

---

# actg320 *ACTG 320 Trial (Second Edition)*

---

## Description

The 1,151-record ACTG 320 HIV treatment-trial dataset described in Table 1.5.

## Format

A data frame with 1,151 rows and 16 variables:

**id** Integer. Participant identifier.

**time** Integer. Follow-up to an AIDS-defining diagnosis or death, or censoring, in days.

**censor** Integer. Composite endpoint indicator: 1 = AIDS-defining diagnosis or death; 0 = censored.

**time_d** Integer. Follow-up to death or censoring, in days.

**censor_d** Integer. Death endpoint indicator: 1 = death; 0 = censored.

**tx** Integer. Treatment indicator: 1 = regimen includes indinavir (IDV); 0 = regimen without IDV.

**txgrp** Integer. Treatment regimen: 1 = ZDV + 3TC; 2 = ZDV + 3TC + IDV; 3 = d4T + 3TC; 4 = d4T + 3TC + IDV. ZDV denotes zidovudine, 3TC lamivudine, and d4T stavudine.

**strat2** Integer. CD4 screening stratum: 0 = CD4 count at most 50; 1 = count above 50.

**sex** Integer. Recorded sex: 1 = male; 2 = female.

**raceth** Integer. Recorded race/ethnicity: 1 = White non-Hispanic; 2 = Black non-Hispanic; 3 = Hispanic, regardless of race; 4 = Asian or Pacific Islander; 5 = American Indian or Alaskan Native; 6 = other/unknown.

**ivdrug** Integer. History of intravenous drug use: 1 = never; 2 = current; 3 = previous.

**hemophil** Integer. Haemophilia indicator: 1 = yes; 0 = no.

**karnof** Integer. Karnofsky Performance Scale: 100 = normal, no complaints or evidence of disease; 90 = normal activity possible with minor signs or symptoms; 80 = normal activity with effort and some signs or symptoms; 70 = self-care possible, but normal activity or active work not possible.

**cd4** Numeric. Baseline CD4 count derived from multiple measurements, in cells/millilitre as stated in the codebook.

**priorzdv** Numeric. Duration of prior zidovudine use, in months.

**age** Integer. Age at enrolment, in years.

## Details

Each row represents a trial participant. The composite AIDS/death endpoint is recorded in `time` and `censor`; the mortality endpoint is recorded in `time_d` and `censor_d`.

`strat2` records the screening CD4 stratum, and `cd4` records the baseline CD4 measurement.

## Download

Use `download_asa("actg320")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

## References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 3, Stata examples. Table 3.2.

## See Also

`actg320ncc`

`vignette("survival-examples", package = "asaData")` for longer analyses.

## Examples

```
## Not run:
actg320 <- download_asa("actg320")
library(survival)
# Second edition, Table 3.2: AIDS-defining diagnosis or death.
actg_cox <- coxph(
  Surv(time, censor == 1) ~ tx + age + sex + cd4 + priorzdv,
  data = actg320, ties = "breslow"
)
summary(actg_cox)

## End(Not run)
```

`actg320ncc` *ACTG 320 Nested Case-Control Sample (Second Edition)*

## Description

Nested case-control data for the ACTG 320 example in Section 9.4, comprising 96 matched sets.

## Format

A data frame with 576 rows and 7 variables:

**set** Integer. Matched case-control set identifier, 1 to 96.

**case** Integer. Status in the sampled set: 1 = case; 0 = control.

**id** Integer. Original ACTG 320 participant identifier; participants may appear in more than one matched set.

**time** Numeric. Follow-up time, in days. The codebook states that case times were jittered to break ties.

**tx** Integer. Treatment indicator: 1 = regimen includes indinavir (IDV); 0 = regimen without IDV.

**age** Integer. Age at enrolment, in years.

**cd4** Numeric. Baseline CD4 count, in cells/millilitre as stated in the codebook.

## Details

Each row represents a sampled case or control. The sample contains 96 matched sets, each with one case and five controls. Participants may appear in more than one set.

`case` identifies cases and controls, and `set` identifies the matched sets. Case follow-up times in `time` are jittered to break ties.

## Download

Use `download_asa("actg320ncc")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

## References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

## See Also

`actg320`

## Examples

```
## Not run:
actg320ncc <- download_asa("actg320ncc")
length(unique(actg320ncc$id))
table(table(actg320ncc$set))

## End(Not run)
```

---

`asa_datasets` *List the Available Applied Survival Analysis Datasets*

---

## Description

List the datasets supported by `download_asa()` without accessing the internet.

## Usage

```
asa_datasets()
```

## Value

A data frame with dataset names, the source edition, dimensions, source filenames, MD5 checksums, and archive URLs. The `ucla_file`, `ucla_md5`, and `ucla_url` columns are empty where no verified UCLA alternative is available. The shared `recur` and `uis` datasets download from the second-edition Wiley archive; `first_edition_md5` identifies their equivalent first-edition files for local imports.

## See Also

`download_asa`, `read_asa`, `read_asa_all`

## Examples

```
asa_datasets()
```

---

bpd *Bronchopulmonary Dysplasia Data (Second Edition)*

---

## Description

Oxygen-use follow-up for 78 subjects in the BPD dataset used in Chapter 2.

## Format

A data frame with 78 rows and 4 variables:

**id** Integer. Subject identifier, 1 to 78.

**surfact** Integer. Surfactant use: 0 = no; 1 = yes.

**ondays** Integer. Time on oxygen, in days, to discontinuation or the end of follow-up.

**censor** Integer. Outcome indicator: 1 = off oxygen; 0 = still on oxygen at study end.

## Details

Each row represents a subject. `ondays` records time on oxygen, and `censor == 1` indicates discontinuation of oxygen.

## Download

Use `download_asa("bpd")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

## References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 2, Stata examples. Figure 2.11 and Table 2.17.

## See Also

`vignette("survival-examples", package = "asaData")` for longer analyses.

## Examples

```
## Not run:
bpd <- download_asa("bpd")
library(survival)
# Second edition, Figure 2.11 and Table 2.17: time to oxygen discontinuation.
bpd_km <- survfit(
  Surv(ondays, censor == 1) ~ surfact, data = bpd
)
plot(bpd_km, col = c("#0072B2", "#D55E00"), lty = 1:2,
     conf.int = FALSE, xlab = "Days on oxygen",
     ylab = "Probability of remaining on oxygen")
legend("topright", c("No surfactant", "Surfactant"),
       col = c("#0072B2", "#D55E00"), lty = 1:2, bty = "n")
bpd_logrank <- survdiff(
  Surv(ondays, censor == 1) ~ surfact, data = bpd
)
bpd_logrank

## End(Not run)
```

comprisk *Competing-Risks Exercise (Second Edition)*

## Description

The 453-record competing-risks exercise dataset from Chapter 9.

## Format

A data frame with 453 rows and 6 variables:

**id** Integer. Subject identifier.

**age** Integer. Age in years.

**gender** Integer. Recorded sex: 0 = male; 1 = female.

**bmi** Numeric. Body mass index, in kg/m^2.

**time** Integer. Follow-up to the recorded event or censoring, in days.

**ev_typ** Integer. Endpoint type: 1 = cardiovascular disease (CVD); 2 = other cause; 0 = censored.

## Details

Each row represents a subject. `time` records follow-up in days. `ev_typ` distinguishes cardiovascular disease (1), other causes (2), and censoring (0).

## Download

Use `download_asa("comprisk")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

## References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

## See Also

`cvdrisk`

## Examples

```
## Not run:
comprisk <- download_asa("comprisk")
table(comprisk$ev_typ)

## End(Not run)
```

cvdrisk *Cardiovascular Competing-Risks Example (Second Edition)*

## Description

The 65-record CVD risk competing-risks example from Section 9.6.

## Format

A data frame with 65 rows and 6 variables:

**id** Integer. Subject identifier.

**age** Integer. Age in years.

**bmi** Numeric. Body mass index, in kg/m^2.

**gender** Integer. Recorded sex: 0 = male; 1 = female.

**time** Numeric. Follow-up to the recorded event or censoring, in days.

**ev_typ** Integer. Endpoint type: 1 = cardiovascular disease (CVD); 2 = other cause; 0 = censored.

## Details

Each row represents a subject. `time` records follow-up in days. `ev_typ` distinguishes cardiovascular disease (1), other causes (2), and censoring (0).

## Download

Use `download_asa("cvdrisk")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

## References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

## See Also

`comprisk`

## Examples

```
## Not run:
cvdrisk <- download_asa("cvdrisk")
table(cvdrisk$ev_typ)

## End(Not run)
```

`download_asa` *Download an Applied Survival Analysis Dataset*

## Description

Download a companion archive from Wiley or UCLA and return the selected dataset as an R data frame.

## Usage

```
download_asa(
  name,
  cache_dir = file.path(tempdir(), "asaData"),
  refresh = FALSE,
  quiet = TRUE,
  source = c("auto", "wiley", "ucla")
)
```

## Arguments

| | |
|---|---|
| `name` | A single dataset name from `asa_datasets()`, for example `"whas500"`. |
| `cache_dir` | Directory for downloaded ZIP archives. The default is a temporary directory for the current R session. Supply a directory of your choice to reuse downloads between sessions. |
| `refresh` | Download the archive again, even if a cached copy exists. The previous copy is retained if downloading or validating the replacement fails. |
| `quiet` | Suppress download progress messages. |
| `source` | `"auto"`, `"wiley"`, or `"ucla"`. The default is `"auto"`: reuse cached archives, preferring Wiley, then try downloading from Wiley followed by UCLA. UCLA is available for six verified datasets only. A message identifies automatic use of UCLA, even when `quiet = TRUE`. |

## Details

Wiley supplies one ZIP archive per edition. UCLA supplies an SPSS archive; verified alternatives are available for `bpd`, `gbcs`, `grace1000`, `uis`, `whas100`, and `whas500`. Other UCLA copies are not substituted because they are absent, incomplete, or have not been verified against Wiley. Only the requested data file is extracted, to temporary storage. Nothing is downloaded when the package is installed or loaded.

Every source file is checked against its recorded checksum, and every returned dataset is checked for dimensions, missing-value counts, and variable types. Changed or damaged files are rejected. With automatic selection, a failed download or rejected file triggers the next available source; errors from both sources are reported if neither succeeds. Use `refresh = TRUE` to bypass cached copies. Each source has a separate cache, and a failed replacement leaves the previous copy intact.

Variable names are lower case, category codes remain numeric, calendar dates become `Date` objects, and missing values become `NA`. UCLA imports use **foreign**. Names and dates are converted to the Wiley definitions, WHAS100 rows are restored to ID order, UIS Beck scores are rounded to three decimal places, and WHAS BMI values are rounded to five decimal places. These conversions reproduce all Wiley values and row orders for the six supported copies. The shared `recur` and

`uis` datasets use the second-edition definitions; the two WHAS nested case-control samples have separate names.

The package contains no dataset records or original codebooks. The GPL-3 licence covers the package code and original documentation, not downloaded materials. Consult the source terms before downloading or reusing the data. For offline use, pass a saved file, ZIP archive, or folder to `read_asa()`.

## Value

A data frame. No objects are assigned to the user's workspace. Download and validation failures produce an informative error.

## Source

Wiley companion data site.

UCLA companion page.

## See Also

`asa_datasets`, `read_asa`, `whas500`

## Examples

```
## Not run:
whas500 <- download_asa("whas500")
head(whas500)
# The second-edition archive is reused for this request.
gbcs <- download_asa("gbcs")
# Select a source explicitly, or keep copies between sessions.
whas500 <- download_asa("whas500", source = "ucla")
whas500 <- download_asa("whas500", cache_dir = "~/asaData-cache")

## End(Not run)
```

---

# `frtcs` *French Three Cities Study (Second Edition)*

---

## Description

Repeated blood-pressure measurements and follow-up for 697 participants in the FRTCS dataset used in Exercise 6 of Chapter 7.

## Format

A data frame with 697 rows and 17 variables:

**id** Integer. Participant identifier, 1 to 697.

**age** Integer. Age at enrolment, in years.

**sex** Integer. Recorded sex: 1 = male; 2 = female.

**date0** Date. Baseline examination date, stored as an R Date.

**sbp0** Numeric. Baseline systolic blood pressure, in mmHg.

**dbp0** Numeric. Baseline diastolic blood pressure, in mmHg.

**antihyp0** Integer. Antihypertensive drug-use indicator in measurement block 0: 0 = no; 1 = yes.

**date1** Date. Examination date in measurement block 1, stored as an R Date.

**sbp1** Numeric. Systolic blood pressure at first follow-up, in mmHg.

**dbp1** Numeric. Diastolic blood pressure at first follow-up, in mmHg.

**antihyp1** Integer. Antihypertensive drug use at first follow-up: 0 = no; 1 = yes.

**date2** Date. Date of the second follow-up examination, stored as an R Date.

**sbp2** Numeric. Systolic blood pressure at second follow-up, in mmHg. Missing values: 3.

**dbp2** Numeric. Diastolic blood pressure at second follow-up, in mmHg. Missing values: 3.

**antihyp2** Integer. Antihypertensive drug use at second follow-up: 0 = no; 1 = yes.

**date_event** Date. Date of the event or end of follow-up, stored as an R Date.

**censor** Integer. Outcome indicator: 1 = event; 0 = censored.

## Details

Each row represents a participant, with repeated blood-pressure measurements in separate columns. Calendar follow-up in days can be calculated as `as.numeric(date_event - date0)`.

## Download

Use `download_asa("frtcs")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

## References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

## Examples

```
## Not run:
frtcs <- download_asa("frtcs")
with(frtcs, summary(as.numeric(date_event - date0)))

## End(Not run)
```

gbcs *German Breast Cancer Study (Second Edition)*

## Description

Recurrence and mortality follow-up for 686 women in the German Breast Cancer Study (GBCS), described in Table 1.4.

## Format

A data frame with 686 rows and 16 variables:

**id** Integer. Participant identifier, 1 to 686.

**diagdate** Date. Diagnosis date, stored as an R Date.

**recdate** Date. Date of recurrence or censoring, stored as an R Date; interpreted with `censrec`.

**deathdate** Date. Date of death or censoring, stored as an R Date; interpreted with `censdead`.

**age** Integer. Age at diagnosis, in years.

**menopause** Integer. Menopausal status: 1 = premenopausal; 2 = postmenopausal.

**hormone** Integer. Hormone therapy: 1 = no; 2 = yes.

**size** Integer. Tumour size, in millimetres.

**grade** Integer. Tumour grade: 1, 2, or 3, corresponding to grades I, II, and III.

**nodes** Integer. Number of involved lymph nodes, from 1 to 51.

**prog_recp** Integer. Progesterone receptor measurement.

**estrg_recp** Integer. Oestrogen receptor measurement.

**rectime** Integer. Time from diagnosis to recurrence or censoring, in days; equals `recdate - diagdate`.

**censrec** Integer. Recurrence indicator: 0 = censored; 1 = recurrence.

**survtime** Integer. Days from diagnosis to death or censoring; equals `deathdate - diagdate`.

**censdead** Integer. Death indicator: 0 = censored; 1 = death.

## Details

Each row represents a woman. The recurrence endpoint is recorded in `rectime` and `censrec`; the mortality endpoint is recorded in `survtime` and `censdead`.

## Download

Use `download_asa("gbcs")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

## References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 4, Stata examples. Table 4.11.

## See Also

`vignette("survival-examples", package = "asaData")` for longer analyses.

## Examples

```
## Not run:
gbcs <- download_asa("gbcs")
library(survival)
# Second edition, Table 4.11: recurrence and hormone-by-size interaction.
# Use a local 0/1 indicator: the source hormone variable is coded 1/2.
gbcs_analysis <- transform(gbcs, hormone01 = as.integer(hormone == 2))
gbcs_cox <- coxph(
  Surv(rectime, censrec == 1) ~ hormone01 * size,
  data = gbcs_analysis, ties = "breslow"
)
coef(gbcs_cox)

## End(Not run)
```

---

# `grace1000` *GRACE Acute Coronary Syndrome Data (Second Edition)*

---

## Description

The 1,000-record GRACE dataset used in Chapter 7 and described in Table 7.4.

## Format

A data frame with 1,000 rows and 9 variables:

**id** Integer. Patient identifier, 1 to 1000.

**days** Numeric. Follow-up to death or censoring, in days, from 0.5 to 180. Same-day deaths or discharges are assigned 0.5 days in the source description.

**death** Integer. Death during follow-up: 1 = death; 0 = censored.

**revasc** Integer. Revascularisation performed: 1 = yes; 0 = no.

**revascdays** Integer. Time from admission, in days: to revascularisation if `revasc` = 1, or to recorded follow-up otherwise.

**los** Integer. Length of hospital stay, in days.

**age** Integer. Age at admission, in years.

**sysbp** Integer. Systolic blood pressure on admission, in mmHg.

**stchange** Integer. ST-segment deviation on the index electrocardiogram: 1 = yes; 0 = no.

## Details

Each row represents a patient. `days` records mortality follow-up, with `death == 1` indicating death. Same-day deaths or discharges are assigned 0.5 days.

For patients with `revasc == 1`, `revascdays` gives the time from admission to revascularisation.

## Download

Use `download_asa("grace1000")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

## References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 7, Stata examples. Table 7.6.

## See Also

`vignette("survival-examples", package = "asaData")` for longer analyses.

## Examples

```
## Not run:
grace1000 <- download_asa("grace1000")
library(survival)
# Second edition, Table 7.6: revascularisation changes during follow-up.
grace_intervals <- tmerge(
  grace1000, grace1000, id = id, death_event = event(days, death)
)
grace_intervals <- tmerge(
  grace_intervals, grace1000[grace1000$revasc == 1, ],
  id = id, revasc_t = tdc(revascdays)
)
grace_cox <- coxph(
  Surv(tstart, tstop, death_event) ~ revasc_t +
    I(1000 / age) + sqrt(sysbp) + stchange,
  data = grace_intervals, ties = "breslow"
)
c(intervals = nrow(grace_intervals), deaths = grace_cox$nevent)
summary(grace_cox)

## End(Not run)
```

`hmohiv` *Hypothetical HMO-HIV Study (First Edition)*

## Description

Follow-up of 100 hypothetical HIV-positive members of a health maintenance organisation, used in Table 1.1 of the first edition.

## Format

A data frame with 100 rows and 7 variables:

**id** Integer. Subject identifier, 1 to 100.

**entdate** Date. Study entry date, stored as an R Date.

**enddate** Date. End-of-follow-up date, stored as an R Date. For an observed death this is the event date; otherwise it is the censoring date.

**time** Integer. Survival or censoring time in approximate months.

**age** Integer. Age in years.

**drug** Integer. History of intravenous drug use: 0 = no; 1 = yes.

**censor** Integer. Follow-up outcome: 1 = death due to AIDS or AIDS-related factors; 0 = alive at study end or lost to follow-up.

## Details

Each row represents a subject. `time` records survival or censoring time in approximate months, and `censor == 1` indicates death due to AIDS or AIDS-related factors. Calendar follow-up in days can be calculated as `as.numeric(enddate - entdate)`.

## Download

Use `download_asa("hmohiv")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

## References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

## Examples

```
## Not run:
hmohiv <- download_asa("hmohiv")
with(hmohiv, table(drug, censor))
with(hmohiv, head(as.numeric(enddate - entdate)))

## End(Not run)
```

read_asa *Read a Locally Downloaded Applied Survival Analysis Dataset*

## Description

Import a saved data file, ZIP archive, or folder without accessing the internet or writing a cache.

## Usage

```
read_asa(name, file)
```

## Arguments

| | |
|---|---|
| name | A single dataset name from asa_datasets(). |
| file | Path to an original Wiley '.dat' file, a supported UCLA '.sav' file, either site's ZIP archive, or a folder containing these files. |

## Details

The same source checks and conversions are used as for download_asa(). Edited files and files with changed line endings are rejected if their checksums differ. Either edition's original recur and uis files is accepted; values agree, and the returned names follow the second edition. Other files must match the requested edition. UCLA support is limited to the six datasets listed in download_asa().

In a folder, the function looks for the expected Wiley filename, then the supported UCLA filename, then an archive named 'edition1.zip' or 'edition2.zip' as appropriate, Wiley's original archive name, 'ucla-asa2-spss.zip', or 'asa2_spss.zip'. The first matching file is checked; an invalid copy is reported rather than silently skipped. Subfolders are not searched. Keep different editions in separate folders because some filenames are shared. A ZIP archive supplied directly may have any filename ending in '.zip'.

## Value

A data frame with the variables described in the corresponding dataset help page.

## See Also

download_asa, asa_datasets, read_asa_all

## Examples

```
## Not run:
whas500 <- read_asa("whas500", "path/to/whas500.dat")
whas500 <- read_asa("whas500", "path/to/Second-Edition Data.zip")
whas500 <- read_asa("whas500", "~/asaData-cache")
whas500 <- read_asa("whas500", "path/to/asa2_spss.zip")

## End(Not run)
```

`read_asa_all` *Read All Available Datasets from Local Archives or Folders*

## Description

Import supported datasets together without accessing the internet or assigning objects to the user's workspace.

## Usage

```
read_asa_all(file)
```

## Arguments

| | |
|---|---|
| `file` | One or more paths to saved ZIP archives or folders containing original data files or ZIP archives. Subfolders are not searched. |

## Details

The first-edition Wiley archive supplies five datasets; the second supplies thirteen. Supply both archives, or a folder containing both, to obtain all 16 datasets. Shared `recur` and `uis` files are represented once. The two WHAS nested case-control samples are distinguished by their checksums. A UCLA SPSS archive supplies the six verified alternatives described in `download_asa()`.

Original filenames inside archives or folders are used to identify candidate data files; archive filenames may be changed. Codebooks and unsupported files are ignored. Every recognised data file must pass its source checksum and data checks, including duplicate copies. A damaged recognised file stops the import with an error. When Wiley and UCLA copies are both supplied, Wiley is used. Files are extracted only to temporary storage and removed after import.

## Value

A named list of data frames, in the order used by `asa_datasets()`. Only datasets available in the supplied paths are included. Use `names()` to see which were imported. An error is raised if no supported data files are found. Optionally use `list2env()` to create separate objects explicitly.

## See Also

`read_asa`, `download_asa`, `asa_datasets`

## Examples

```
## Not run:
# Both Wiley ZIP archives saved in one folder: returns all 16 datasets.
datasets <- read_asa_all("~/Downloads/asa-data")
names(datasets)
# Alternatively, supply the archive paths directly.
datasets <- read_asa_all(c("path/to/First-Edition Data.zip",
                           "path/to/Second-Edition Data.zip"))
# Optional: create a separate R object for each dataset.
list2env(datasets, envir = .GlobalEnv)

## End(Not run)
```

`recur` *Recurrent Soreness Episodes*

## Description

Counting-process data from a hypothetical study of recurrent soreness episodes, used in the Chapter 9 exercises.

## Format

A data frame with 1,296 rows and 7 variables:

**id** Integer. Subject identifier, 1 to 400; repeated across intervals.

**age** Integer. Age in years.

**treat** Integer. Treatment assignment: 0 = new treatment; 1 = old treatment.

**time0** Integer. Start of the risk interval, measured in days from the study time origin; the time of the previous episode, or zero for the first interval.

**time1** Integer. End of the risk interval, measured in days from the same origin; the time of the next episode or censoring.

**censor** Integer. Interval outcome: 1 = a soreness episode occurred at time1; 0 = censored at time1.

**event** Integer. Episode or risk-interval number. The supplied values are 1 to 4; the codebook gives a possible range of 0 to 4.

## Details

Each row represents a risk interval for one subject. There are 1,296 intervals from 400 subjects, with up to four intervals per subject. The interval is (`time0`, `time1`], and `censor == 1` indicates an episode at its end.

The first- and second-edition files contain identical values and variable names. `recur` represents both editions using the second-edition file. The second-edition codebook specifies that `censor = 1` denotes an episode at `time1`.

## Download

Use `download_asa("recur")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

## References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

## Examples

```
## Not run:
recur <- download_asa("recur")
length(unique(recur$id))
head(recur[order(recur$id, recur$time0), ])

## End(Not run)
```

---

`uis` *UMass Impact Study of Drug Treatment*

---

## Description

Drug-treatment follow-up data for 628 participants in the UMass Impact Study (UIS), described in Table 1.3.

## Format

A data frame with 628 rows and 12 variables:

**id** Integer. Participant identifier, 1 to 628.

**age** Integer. Age at enrolment, in years. Missing values: 5.

**beck** Numeric. Beck Depression Score at admission, from 0 to 54. Missing values: 33.

**hercoc** Integer. Heroin/cocaine use during the three months before admission: 1 = both heroin and cocaine; 2 = heroin only; 3 = cocaine only; 4 = neither. Missing values: 18.

**ivhx** Integer. History of intravenous drug use at admission: 1 = never; 2 = previous; 3 = recent. Missing values: 18.

**ndrugtx** Integer. Number of previous drug treatments, from 0 to 40. Missing values: 17.

**race** Integer. Recorded race: 0 = White; 1 = other, using the terminology of the second-edition codebook. Missing values: 6.

**treat** Integer. Randomised treatment assignment: 0 = short programme; 1 = long programme.

**site** Integer. Treatment site: 0 = A; 1 = B.

**los** Integer. Length of stay in treatment, in days from admission to treatment exit.

**time** Integer. Time from admission to the recorded drug-use endpoint or end of follow-up, in days.

**censor** Integer. Event indicator: 1 = returned to drug use; 0 = otherwise, following the second-edition codebook. See the edition comparison below.

## Details

Each row represents a participant. There are 628 records, of which 575 have complete data.

The two editions contain identical values in the same row and column order. `uis` represents both editions using second-edition variable names. The depression-score column is `becktota` in the first edition and `beck` in the second; this dataset uses `beck`.

The first-edition codebook defines `censor = 1` as return to drug use or loss to follow-up. The second-edition codebook describes the same indicator as return to drug use. For `race = 1`, the first edition uses the label non-White, and the second uses other.

## Download

Use `download_asa("uis")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

## References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 4, Stata examples. Table 4.12.

## See Also

`vignette("survival-examples", package = "asaData")` for longer analyses.

## Examples

```
## Not run:
uis <- download_asa("uis")
library(survival)
# Second edition, Table 4.12: common sample for the drug-history models.
uis_analysis <- uis[complete.cases(uis[c("age", "ivhx")]), ]
uis_analysis$drug <- as.integer(uis_analysis$ivhx %in% c(2, 3))
uis_cox <- coxph(
  Surv(time, censor == 1) ~ drug,
  data = uis_analysis, ties = "breslow"
)
c(participants = uis_cox$n, events = uis_cox$nevent)
summary(uis_cox)

## End(Not run)
```

---

# whas *Worcester Heart Attack Study (First Edition)*

---

## Description

The 481-record Worcester Heart Attack Study (WHAS) dataset described in Table 1.4 of the first edition.

## Format

A data frame with 481 rows and 14 variables:

**id** Integer. Patient identifier, 1 to 481.

**age** Integer. Age recorded in the chart, in years.

**sex** Integer. Recorded sex: 0 = male; 1 = female.

**cpk** Integer. Peak cardiac enzyme measurement, in international units as stated in the source codebook.

**sho** Integer. Cardiogenic shock complication: 0 = no; 1 = yes.

**chf** Integer. Left heart failure complication: 0 = no; 1 = yes.

**miord** Integer. Myocardial infarction order: 0 = first; 1 = recurrent.

**mitype** Integer. Myocardial infarction type: 1 = Q-wave; 2 = non-Q-wave; 3 = indeterminate.

**year** Integer. Cohort-year code: 1 = 1975; 2 = 1978; 3 = 1981; 4 = 1984; 5 = 1986; 6 = 1988.

**yrgrp** Integer. Grouped cohort-year code: 1 = 1975 and 1978; 2 = 1981 and 1984; 3 = 1986 and 1988.

**lenstay** Integer. Length of hospital stay, in days.

**dstat** Integer. Status at hospital discharge: 0 = alive; 1 = dead.

**lenfol** Integer. Total follow-up from hospital admission to death or last follow-up, in days.

**fstat** Integer. Vital status at last follow-up: 0 = alive; 1 = dead.

## Details

Each row represents a patient. `lenfol` and `fstat` record mortality follow-up; `lenstay` and `dstat` record the hospital-discharge endpoint.

The cohort and variable coding differ from `whas100` and `whas500`. In particular, `mitype` has three categories here and two in `whas500`.

## Download

Use `download_asa("whas")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

## References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

## See Also

`whas100`, `whas500`, `whasncc_e1`

## Examples

```
## Not run:
whas <- download_asa("whas")
str(whas)
head(whas)

## End(Not run)
```

whas100 *WHAS 100-Patient Dataset (Second Edition)*

## Description

The 100-patient Worcester Heart Attack Study dataset described in Table 1.1.

## Format

A data frame with 100 rows and 9 variables:

**id** Integer. Patient identifier, 1 to 100.

**admitdate** Date. Hospital admission date, stored as an R Date.

**foldate** Date. Date of death or last follow-up, stored as an R Date.

**los** Integer. Length of hospital stay, in days.

**lenfol** Integer. Follow-up from hospital admission to death or last follow-up, in days.

**fstat** Integer. Vital status at last follow-up: 1 = dead; 0 = alive.

**age** Integer. Age at admission, in years.

**gender** Integer. Recorded sex: 0 = male; 1 = female.

**bmi** Numeric. Body mass index, in kg/m^2.

## Details

Each row represents a patient. `lenfol` records mortality follow-up, with `fstat == 1` indicating death.

## Download

Use `download_asa("whas100")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

## References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 2, Stata examples. Table 2.3 and Figure 2.2.

## See Also

`whas`, `whas500`

`vignette("survival-examples", package = "asaData")` for longer analyses.

## Examples

```
## Not run:
whas100 <- download_asa("whas100")
library(survival)
# Second edition, Table 2.3 and Figure 2.2: all-cause mortality.
whas100_km <- survfit(
  Surv(lenfol, fstat == 1) ~ 1,
  data = whas100, conf.type = "log-log"
)
summary(whas100_km, times = c(6, 14, 44))
plot(whas100_km, xscale = 365.25, conf.int = FALSE,
     xlab = "Years since admission", ylab = "Survival probability")

## End(Not run)
```

---

`whas500` *WHAS 500-Patient Dataset (Second Edition)*

---

## Description

The 500-patient Worcester Heart Attack Study dataset described in Table 1.2.

## Format

A data frame with 500 rows and 22 variables:

**id** Integer. Patient identifier, 1 to 500.

**age** Integer. Age at hospital admission, in years.

**gender** Integer. Recorded sex: 0 = male; 1 = female.

**hr** Integer. Initial heart rate, in beats per minute.

**sysbp** Integer. Initial systolic blood pressure, in mmHg.

**diasbp** Integer. Initial diastolic blood pressure, in mmHg.

**bmi** Numeric. Body mass index, in kg/m^2.

**cvd** Integer. History of cardiovascular disease: 0 = no; 1 = yes.

**afb** Integer. Atrial fibrillation: 0 = no; 1 = yes.

**sho** Integer. Cardiogenic shock: 0 = no; 1 = yes.

**chf** Integer. Congestive heart failure complication: 0 = no; 1 = yes.

**av3** Integer. Complete heart block: 0 = no; 1 = yes.

**miord** Integer. Myocardial infarction order: 0 = first; 1 = recurrent.

**mitype** Integer. Myocardial infarction type: 0 = non-Q-wave; 1 = Q-wave.

**year** Integer. Cohort-year code: 1 = 1997; 2 = 1999; 3 = 2001.

**admitdate** Date. Hospital admission date, stored as an R Date.

**disdate** Date. Hospital discharge date, stored as an R Date.

**fdate** Date. Date of death or last follow-up, stored as an R Date.

**los** Integer. Length of hospital stay, in days, from admission to discharge.

**dstat** Integer. Hospital discharge status: 0 = alive; 1 = dead.

**lenfol** Integer. Total follow-up from admission to death or last follow-up, in days.

**fstat** Integer. Vital status at last follow-up: 0 = alive; 1 = dead.

## Details

Each row represents a patient. `lenfol` and `fstat` record mortality follow-up; `los` and `dstat` record the hospital-discharge endpoint.

The coding of `mitype` and `year` differs from that in `whas`.

## Download

Use `download_asa("whas500")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

## References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 5, Stata examples. Table 5.3.

## See Also

`whas`, `whas100`, `whasncc_e2`

`vignette("survival-examples", package = "asaData")` for longer analyses.

## Examples

```
## Not run:
whas500 <- download_asa("whas500")
library(survival)
# Second edition, Table 5.3: all-cause mortality after admission.
whas500_cox <- coxph(
  Surv(lenfol, fstat == 1) ~ age + hr + sysbp + diasbp + bmi +
    gender + cvd + afb + chf + miord + mitype,
  data = whas500, ties = "breslow"
)
summary(whas500_cox)

## End(Not run)
```

---

# whasncc_e1 *WHAS Nested Case-Control Sample (First Edition)*

---

## Description

Nested case-control data for the Worcester Heart Attack Study, used in the Chapter 9 exercises.

## Format

A data frame with 1,494 rows and 12 variables:

**set** Integer. Matched case-control set identifier.

**case** Integer. Status within this sampled set: 0 = control; 1 = case.

**t** Numeric. Follow-up time with ties broken, in days.

**lenfol** Integer. Total follow-up from hospital admission, in days.

**fstat** Integer. Vital status at last follow-up in the cohort: 0 = alive; 1 = dead.

**age** Integer. Age recorded in the chart, in years.

**sex** Integer. Recorded sex: 0 = male; 1 = female.

**chf** Integer. Left heart failure complication: 0 = no; 1 = yes.

**miord** Integer. Myocardial infarction order: 0 = first; 1 = recurrent.

**yrgrp** Integer. Grouped cohort-year code: 1 = 1975 and 1978; 2 = 1981 and 1984; 3 = 1986 and 1988.

**lenstay** Integer. Length of hospital stay, in days.

**nr** Integer. Size of the underlying cohort risk set at sampling. Missing values: 858.

## Details

Each row represents a sampled case or control within a risk set. There are 249 matched sets, each with one case and five controls. Individuals may appear in multiple risk sets.

`case` identifies cases and controls, and `set` identifies the matched sets. `fstat` records vital status in the underlying cohort.

The editions contain different sampled records and variables. The first edition has 1,494 rows in 249 matched sets and 12 variables; the second has 1,638 rows in 273 sets and 11 variables. The second edition includes `bmi` but omits `yrgrp` and `lenstay`.

## Download

Use `download_asa("whasncc_e1")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

## References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

## See Also

`whasncc_e2`

## Examples

```
## Not run:
whasncc_e1 <- download_asa("whasncc_e1")
table(table(whasncc_e1$set))
with(whasncc_e1, table(case, fstat))

## End(Not run)
```

whasncc_e2 *WHAS Nested Case-Control Sample (Second Edition)*

## Description

Nested case-control data for the Worcester Heart Attack Study, used in the Chapter 9 exercises.

## Format

A data frame with 1,638 rows and 11 variables:

**set** Integer. Matched case-control set identifier.

**case** Integer. Status within this sampled set: 0 = control; 1 = case.

**t** Numeric. Follow-up time with ties broken, in days.

**lenfol** Integer. Total follow-up from hospital admission, in days.

**fstat** Integer. Vital status at last follow-up in the cohort: 0 = alive; 1 = dead.

**age** Integer. Age recorded in the chart, in years.

**sex** Integer. Recorded sex: 0 = male; 1 = female.

**bmi** Numeric. Body mass index, in kg/m^2.

**chf** Integer. Left heart failure complication: 0 = no; 1 = yes.

**miord** Integer. Myocardial infarction order: 0 = first; 1 = recurrent.

**nr** Integer. Size of the underlying cohort risk set at sampling. Missing values: 1365.

## Details

Each row represents a sampled case or control within a risk set. There are 273 matched sets, each with one case and five controls. Individuals may appear in multiple risk sets.

`case` identifies cases and controls, and `set` identifies the matched sets. `fstat` records vital status in the underlying cohort.

The editions contain different sampled records and variables. The first edition has 1,494 rows in 249 matched sets and 12 variables; the second has 1,638 rows in 273 sets and 11 variables. The second edition includes `bmi` but omits `yrgrp` and `lenstay`.

## Download

Use `download_asa("whasncc_e2")` to retrieve this dataset. Data are not included in the package.

## Source

Wiley companion data site.

## References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

## See Also

whasncc_e1

## Examples

```
## Not run:
whasncc_e2 <- download_asa("whasncc_e2")
table(table(whasncc_e2$set))
with(whasncc_e2, table(case, fstat))

## End(Not run)
```

# Index


∗ **datasets**
actg320, 3
actg320ncc, 5
asaData-package, 2
bpd, 6
comprisk, 8
cvdrisk, 9
frtcs, 11
gbcs, 13
grace1000, 14
hmohiv, 16
recur, 19
uis, 20
whas, 21
whas100, 23
whas500, 24
whasncc_e1, 25
whasncc_e2, 27
∗ **package**
asaData-package, 2

actg320, 3, *5*
actg320ncc, *4*, 5
asa_datasets, *2, 3*, 6, *11*, *17, 18*
asaData *(*asaData-package*)*, 2
asaData-package, 2

bpd, 6

comprisk, 8, *9*
cvdrisk, *8*, 9

download_asa, *2, 3*, *6*, 10, *17, 18*

frtcs, 11

gbcs, 13
grace1000, 14

hmohiv, 16

read_asa, *2, 3*, *6*, *11*, 17, *18*
read_asa_all, *2*, *6*, *17*, 18
recur, *2*, 19

uis, *2*, 20

whas, 21, *23*, *25*
whas100, *22*, 23, *25*
whas500, *11*, *22, 23*, 24
whasncc_e1, *2*, *22*, 25, *27*
whasncc_e2, *2*, *25, 26*, 27